



MindCap

Research & Validation Roadmap

Version 1.0 | February 2024

Timeline: 3-4 months to validated MVP

Executive Summary

You are currently in the **research and validation phase**, not ready to launch. This roadmap provides a structured 16-week plan to validate your core algorithms before considering a public launch.

Key Insight: You thought you were at 'polish and launch' but you're actually at 'validate core algorithms.' This is good—you caught it before launching something broken.

Current Reality Check

What You Have:

- Extension infrastructure (Plasmo, IndexedDB, sync)
- Backend architecture (FastAPI, Supabase)
- Privacy-first design
- Clear differentiation (rabbit holes as features)
- 56,000 Firefox history records for testing

What You Need:

- Validated keyword extraction (80%+ accuracy)
- Validated session detection (90%+ accuracy)
- Validated rabbit hole classification (75%+ accuracy)
- Working topic chain reconstruction
- Clear, compelling visualization
- Real user feedback (5+ people saying 'I'd use this')

Phase 1: Algorithm Development (Weeks 1-6)

Week 1-2: Keyword Extraction Validation

Goal: 80%+ accuracy on extracting topics from page titles

- Create validation notebook with 100-sample test set
- Port TypeScript algorithm to Python for testing
- Manual review of all 100 extractions
- Calculate accuracy metrics
- Identify failure patterns
- Iterate on algorithm

Common Issues You'll Find:

- Domain names appearing as keywords (github, reddit)
- Generic words not filtered (tutorial, guide, best)
- Missing compound terms (machine learning → machine, learning)
- Comparison patterns not detected (X vs Y)

Success Criteria:

$$\text{Accuracy} = (\text{Good} + \text{Okay}) / \text{Total} \geq 80\%$$

Where Good = captures right topics, Okay = acceptable, Bad = wrong

Time Allocation:

Day	Activity
1-2	Set up notebook, run extraction
3-4	Manual review (50 samples per day)
5-6	Analysis and iteration
7	Update extension code

Week 3-4: Session Detection Validation

Goal: 90%+ correct session boundaries

Daily Browsing Scenarios:

Monday: Deep Research: 2-hour focused research, test if coffee break splits session

Tuesday: Fragmented Browsing: Multiple short sessions throughout day, test time gaps

Wednesday: Rabbit Hole: Start with one topic, end elsewhere, test topic drift

Thursday: Comparison Shopping: Research products, test back-and-forth navigation

Friday: Work + Interruption: Focused work interrupted by distraction, test context switch

Metrics to Track:

Metric	Formula	Target
Boundary Accuracy	Correct boundaries / Total	90%+
Type Accuracy	Correct type / Total	70%+
False Splits	Incorrectly split / Total	<10%
False Merges	Incorrectly merged / Total	<5%

Parameters to Tune:

maxGapMinutes: 30 (tune: 20? 30? 45?)

topicShiftThreshold: 0.2 (keyword overlap)

videoGapTolerance: 60 (longer gaps during video)

Week 5-6: Rabbit Hole Classification

Goal: 75%+ agreement between manual and algorithm classification

The 4 Rabbit Hole Types:

Type	Characteristics	Example
Coherent Deep Dive	High depth, low spread, high coherence	kubernetes → networking → TCP → OSI
Wandering Journey	High depth, high spread, logical steps	hooks → redux → event sourcing → DDD
Tangent Hopper	Random jumps, "how did I get here?"	docker → pizza → renaissance → crypto
Focused Exploration	Low depth, high branching	kubernetes → (docs, video, reddit, SO)

Activities:

- Collect 20 real rabbit hole sessions from your browsing
- Manually classify each before looking at algorithm output
- Compare manual vs algorithm classification
- Create confusion matrix
- Tune classification thresholds
- Document each type clearly

Phase 2: Visualization Development (Weeks 9-12)

Week 9-10: Design Explorations

Now that your data is good, how do you show it? Explore 4 visualization options:

Linear Timeline (RECOMMENDED): Simple, easy to understand, mobile-friendly

Tree Diagram (For complex cases): Shows branching clearly

Sankey Diagram (Advanced): Shows time + flow

Network Graph (Post-MVP): Shows connections

Your Task:

1. Mock up all 4 options (Figma or hand-drawn)
2. Test with your own rabbit hole data
3. Ask: 'Which one makes me go aha!'?
4. Pick ONE for MVP

Week 11-12: Build One Visualization

Pick ONE visualization and build it properly. Don't build all 4—that's scope creep.

Iterate Until:

- You can understand the flow at a glance
- The visualization reveals something interesting
- You'd want to share it with someone

Phase 3: Self-Dogfooding (Weeks 13-16)

Week 13-14: Use It Daily

Goal: Live with your own tool for 2 weeks

Daily Routine:

- Browse normally (don't change behavior)
- Check dashboard once per day
- Keep a journal of observations

Red Flags:

- You stop checking after 3 days → Core value isn't there
- You check but don't learn anything → Insights are weak
- You're embarrassed by rabbit holes → Framing is wrong

Green Flags:

- You check daily without reminders
- You screenshot rabbit holes to share
- You learn something about yourself
- You want to show it to friends

Week 15-16: Show 5 Friends

Goal: Validate with real users

Pick 5 People Who:

- Browse a lot (knowledge workers, students, researchers)
- Are comfortable with early software
- Will give honest feedback

Process:

1. Install for them (sit with them, don't just send link)
2. Have them browse for 3 days
3. Review dashboard together
4. Ask open-ended questions

Decision Point (Week 16)

Success Criteria:

■ **GO if:** 3 out of 5 find it interesting (7+ on 10-point scale), 2 out of 5 would pay something, 1 out of 5 asks 'when can I use this?'

■ **NO-GO if:** Less than 50% find it interesting, less than 30% would pay, fundamental confusion about concept

Option A: You Have Something (3+ friends loved it)

Next Steps:

1. Polish the MVP (2 weeks)
2. Recruit 20 beta users (1 week)
3. Launch in 6 weeks

Timeline to Launch: 9 weeks from decision point

Option B: Interesting But Not There Yet (1-2 friends loved it)

Next Steps:

1. Identify the gap (what's missing?)
2. Build that one thing (4 weeks)
3. Test with 5 more friends (2 weeks)
4. Re-evaluate

Timeline to Launch: 12-16 weeks from decision point

Option C: Nobody Cares (Less than 1 friend loved it)

Pivot Options:

- Different framing (not 'rabbit holes' but 'research patterns')
- Different audience (ADHD community, researchers, enterprise)
- Different product (weekly email, real-time nudges)
- Kill it (take learnings to new problem)

Success Metrics Summary

Phase	Week	Milestone	Success Criteria
1	2	Keyword extraction	80%+ accuracy
1	4	Session detection	90%+ boundaries
1	6	Rabbit hole classification	75%+ types
1	8	Topic chains	100% tree structure
2	10	Visualization	"Whoa" reaction
2	12	Dashboard	End-to-end works
3	14	Self-dogfooding	Daily use, interesting
3	16	User validation	3/5 love it

Critical Path Dependencies

Each phase depends on the previous. You cannot skip steps:

Keyword Extraction (2w) →

Session Detection (2w) →

Rabbit Hole Classification (2w) →

Topic Chain Reconstruction (2w) →

Visualization (2w) →

Dashboard (2w) →

Self-Dogfooding (2w) →

User Validation (2w) →

GO/NO-GO DECISION

What NOT to Do

■ Don't build the dashboard first

You'll have nothing to show in it. Waste time on UI when algorithms are broken.

■ Don't perfect one thing for months

80% accuracy is good enough. Diminishing returns after that.

■ Don't build all 4 visualizations

Pick one, validate it works. Add others only if users ask.

■ Don't add new features

No account linking, mobile app, or other patterns yet. Just rabbit holes, done well.

Parallel Work (What You CAN Do)

While working on algorithms, you can:

- Build in public (tweet progress, share findings)
- Design mockups (Figma doesn't require working code)
- Write documentation (privacy policy, FAQ)
- Plan marketing (landing page copy, Product Hunt assets)

But don't build code for these yet. Focus on algorithms first.

The Honest Timeline

Scenario	Timeline	Outcome
Optimistic (everything works first try)	16 weeks to beta	Ready to launch
Realistic (normal iteration)	20-24 weeks to beta	Ready to launch
Pessimistic (major pivots)	28+ weeks or never	Pivot or kill

Remember: You're doing research, not just engineering. Research takes time.

Your Immediate Action Plan

Timeline	Action
TODAY	Set up validation notebook
TODAY	Run extraction on 100 samples
TODAY	Export review CSV
TOMORROW	Manual review (50 samples)
DAY 3	Manual review (50 samples)
DAY 4	Run analysis, identify improvements
DAY 5	Re-run extraction, check if 80%+
WEEKEND	Update extension code, test edge cases
NEXT WEEK	Move to session detection

Final Advice

You're in the **research phase**, not the **shipping phase**. Take time to validate. Iterate until it works. Document everything. Be willing to pivot. Don't rush to ship. Don't skip validation. Don't build UI first.

The goal: Validate that the core idea works before you invest months building a product nobody wants.